PixelPush

Turn Your GitHub Graph Into Art

Product Documentation

Version 1.0

Developed by

Aepostrophee

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1. Introduction

PixelPush is a web-based Progressive Web App (PWA) designed for developers who want to transform their GitHub contribution graph into pixel art. By connecting their GitHub account, users can select a visual theme, generate a pixel art image mapped to the graph's green shading system, and receive a precise commit schedule that guides them to recreate that image on their live GitHub profile — one contribution at a time.

This document covers the complete product specification, technical architecture, feature set, API integrations, and development roadmap for PixelPush, built and maintained by Aepostrophee.

2. Product Overview

2.1 Vision

Every developer has a GitHub contribution graph. Most ignore it. PixelPush turns it into a canvas — making the act of writing code a creative, gamified, and socially shareable experience.

2.2 Core Value Proposition

- Transforms a developer's existing GitHub profile into interactive pixel art
- Removes the complexity of manually planning contribution patterns
- Provides a guided, day-by-day commit schedule to complete artwork
- Creates shareable progress milestones that market the app organically

2.3 Target Audience

Segment	Description
Primary	Software developers active on GitHub with consistent contribution history
Secondary	Open source contributors seeking creative profile differentiation
Tertiary	Developer communities, bootcamp students, and tech content creators

2.4 App Name & Branding

App Name
PixelPush

Tagline
Turn Your GitHub Graph Into Art

Developed By
Aepostrophee

Platform
Progressive Web App (PWA) — accessible on desktop and mobile browsers

3. Feature Specification

3.1 GitHub OAuth Integration

Users connect their GitHub account via OAuth 2.0. The app requests the minimum required scopes to read contribution data and guide the user through future commits.

OAuth Scope	Purpose
read:user	Access basic profile information (username, avatar)
repo	Required to read contribution data across public and private repos
user:email	For account identification and notification delivery

3.2 Contribution Graph Reading

Once authenticated, PixelPush queries GitHub's GraphQL API to retrieve the user's full contribution calendar. The API returns contribution data for each day of the past year including:

- Date of contribution
- Exact contribution count
- Hex color code (one of 5 shade levels used by GitHub)

The five shade levels correspond to:

Hex Color	Shade Level
#ebedf0	Level 0 — No contributions (empty)
#9be9a8	Level 1 — Low contributions (light green)
#40c463	Level 2 — Moderate contributions (mid green)
#30a14e	Level 3 — High contributions (dark green)
#216e39	Level 4 — Very high contributions (darkest green)

Important: GitHub's shading thresholds are relative to each user's personal contribution history, not fixed numbers. PixelPush calculates a per-user threshold baseline from historical data to generate accurate future commit targets.

3.3 Theme Selection & Pixel Art Generation

Users select a visual theme for their pixel art. PixelPush's generation engine maps theme imagery to the 5 available shade levels of the GitHub graph, producing a viable pixel art design within the 52 columns × 7 rows resolution of the contribution grid.

Available Themes (v1.0)

- Aviation — aircraft silhouettes, propellers, flight paths
- Pets — cats, dogs, paw prints, fish bowls
- Cars — classic vehicles, racing cars, tire tracks
- Scenery — mountains, trees, cityscapes, sunsets
- Space — rockets, planets, stars, astronauts
- Abstract — geometric patterns, waves, symbols

For each theme, the app generates 3 to 5 pixel art variations. Each is previewed as a grid overlay matching the exact dimensions of the user's GitHub graph. The user selects one design to proceed with.

3.4 Commit Schedule Generator

After a design is selected, PixelPush calculates a forward-looking commit schedule. This schedule tells the user:

- Which days require contributions
- How many commits to make on each active day
- At what intervals to space commits for natural-looking graph shading

The schedule is displayed as a calendar view with daily tasks. Users can export the schedule or receive it as daily reminders via browser push notifications or email.

Schedule Recovery

If a user misses a day, PixelPush offers a 'Recalculate from Today' feature that regenerates the remaining schedule dynamically, adjusting future commit targets to compensate for missed days without losing the overall design.

3.5 Progress Tracker

PixelPush continuously monitors the user's GitHub contribution graph and compares it against the target design. The progress tracker displays:

- Overall completion percentage
- A live side-by-side view of target design vs. current graph
- Upcoming commit tasks for the next 7 days
- Streak tracking — consecutive days on schedule

3.6 Social Sharing

Sharing is the app's primary organic marketing channel. PixelPush generates dynamic image cards for three key moments in a user's journey:

Sharing Moment	Visual Content
Design Started	Target pixel art overlaid on the user's current contribution graph
Progress Milestone	Side-by-side comparison at 25%, 50%, and 75% completion
Design Completed	Before/after split showing empty graph vs. completed artwork
Timelapse GIF	Animated weekly progression of the graph filling in over time

Sharing Channels

- Twitter / X — pre-filled tweet with generated image card and app URL
- LinkedIn — milestone completion posts for professional visibility
- Web Share API — universal share sheet covering WhatsApp, Telegram, iMessage, and all installed apps

Every share links to a public profile page at pixelpush.app/u/{username} which displays the user's in-progress or completed design with a prominent 'Create Yours' call-to-action. This forms the viral growth loop.

4. Technical Architecture

4.1 Technology Stack

Layer	Technology
Frontend Framework	Next.js 14 (App Router)
Styling	Tailwind CSS
Authentication	NextAuth.js with GitHub OAuth 2.0 provider
GitHub Data	GitHub GraphQL API v4
Pixel Art Rendering	HTML5 Canvas API / D3.js
Image Card Generation	Satori + resvg-js (server-side OG image rendering)
GIF Generation	gifencoder (Node.js, server-side)
Database	PostgreSQL via Prisma ORM
Push Notifications	Web Push API (VAPID)
Hosting & Deployment	Vercel (frontend + API routes)
CDN & Storage	Vercel Blob Storage for generated image cards

4.2 System Architecture Overview

PixelPush follows a serverless architecture pattern, with Next.js API routes handling all backend logic. This eliminates the need for a separate backend server and keeps the infrastructure simple and cost-effective during the early growth phase.

Core Data Flow

1. User lands on PixelPush and clicks 'Connect GitHub'
2. GitHub OAuth flow initiates — user authorizes the app
3. NextAuth.js stores the access token securely in a server-side session
4. PixelPush calls GitHub GraphQL API to fetch the full contribution calendar
5. The contribution grid (52 × 7 days) is parsed, normalized, and stored
6. User selects a theme — the pixel art engine generates design candidates
7. User picks a design — the schedule generator produces a daily commit plan
8. Progress tracker runs on a cron schedule (daily) to update completion status
9. Share cards are generated on demand and cached in Vercel Blob Storage

4.3 GitHub GraphQL API Query

The following GraphQL query is used to retrieve the full contribution calendar for the authenticated user:

```
query GetContributionGraph($from: DateTime, $to: DateTime) {
  user(login: $username) {
    contributionsCollection(from: $from, to: $to) {
      contributionCalendar {
        weeks {
          contributionDays {
            date
            contributionCount
            color
          }
        }
      }
    }
  }
}
```

4.4 Pixel Art Engine

The pixel art generation engine converts theme images into valid GitHub graph art through the following pipeline:

10. Source image is downsampled to 52 × 7 pixels (matching the contribution grid)
11. Each pixel is quantized to one of 5 brightness levels
12. Brightness levels are mapped to GitHub's 5 shade hex values
13. Multiple design variations are generated using different cropping and style parameters
14. Each variation is rendered as an interactive grid preview using the HTML5 Canvas API

4.5 Share Card Generation

Share image cards are generated server-side using Satori (JSX to SVG) combined with resvg-js (SVG to PNG). Cards are generated on demand and cached to Vercel Blob Storage with a unique URL.

The share card API endpoint accepts:

- username — GitHub username
- stage — started / progress / completed
- percent — completion percentage (for progress cards)

Generated cards are returned as PNG images optimized for Twitter (1200 × 630px) and are cached for 24 hours.

5. User Flows

5.1 Onboarding Flow

15. User visits pixelpush.app
16. Clicks 'Connect GitHub' — redirected to GitHub OAuth authorization
17. Grants permission — redirected back to PixelPush dashboard
18. App displays current contribution graph as a pixel grid
19. User is prompted to select a theme category

5.2 Design Creation Flow

20. User selects a theme (e.g., Pets)
21. App generates 3 to 5 pixel art variations for that theme
22. Variations are displayed as overlays on the user's graph dimensions
23. User selects a preferred design
24. App calculates the full commit schedule and displays it as a calendar
25. User confirms and activates the schedule

5.3 Active Schedule Flow

26. User receives daily push notification or email reminder
27. Notification specifies how many commits to make that day
28. User opens app to view today's task and upcoming schedule
29. After making commits, app auto-syncs with GitHub API and updates progress
30. At milestone completions, app prompts user to share progress

5.4 Sharing Flow

31. User taps 'Share Progress' at any milestone
32. App generates a dynamic share card image server-side
33. User selects platform: Twitter, LinkedIn, or universal share
34. Pre-filled post opens with the generated image card and app URL
35. Recipients land on the user's public profile page at /u/{username}
36. Public profile page shows the in-progress design with a 'Create Yours' CTA

6. Database Schema

PixelPush uses PostgreSQL with Prisma ORM. The core tables are:

6.1 Users

Column	Description
id	UUID — primary key
github_id	Integer — GitHub user ID
username	String — GitHub username
avatar_url	String — GitHub avatar URL
access_token	String — encrypted OAuth access token
created_at	Timestamp — account creation date
updated_at	Timestamp — last profile sync

6.2 Designs

Column	Description
id	UUID — primary key
user_id	UUID — foreign key to Users
theme	String — selected theme category
pixel_data	JSONB — 52x7 grid of shade level values
status	Enum — active / completed / abandoned
started_at	Timestamp — design activation date
target_end_at	Timestamp — projected completion date
completed_at	Timestamp — actual completion date (nullable)

6.3 Schedule Entries

Column	Description
id	UUID — primary key
design_id	UUID — foreign key to Designs
date	Date — target contribution date

target_count	Integer — number of commits required
actual_count	Integer — actual commits made (from GitHub sync)
status	Enum — pending / completed / missed / skipped

7. Notifications & Reminders

7.1 Push Notifications

PixelPush implements the Web Push API using VAPID keys for browser-based push notifications. Notifications are sent via a daily cron job that runs at 8:00 AM in the user's local timezone.

Notification Type	Trigger
Daily Reminder	Sent on days the user has a scheduled commit target
Streak Alert	Sent when a user is at risk of breaking a streak (commits due by midnight)
Milestone	Sent when user crosses 25%, 50%, 75%, and 100% completion
Completion	Sent when the full design is finished — includes share prompt

7.2 Email Notifications

As a fallback for users who decline push notifications, PixelPush sends daily schedule digests and milestone alerts via email. Emails are delivered through a transactional email provider (Resend or Postmark).

8. Development Roadmap

Phase 1 — Core MVP

- GitHub OAuth login and contribution graph reading
- Pixel art generation engine with 3 themes
- Commit schedule generator and calendar view
- Basic progress tracker
- Public profile page (/u/username)

Phase 2 — Engagement & Sharing

- Social share card generation (Satori + resvg)
- Web Share API integration
- Twitter / X deep link sharing
- Push notification system
- Streak tracking and badges

Phase 3 — Growth Features

- Timelapse GIF generator for completed designs
- Expanded theme library (10+ themes)
- Schedule recalculation (recovery from missed days)
- Email notification fallback
- Community gallery of completed designs

Phase 4 — Monetization

- Launch PixelPush Pro subscription at $6-8/month
- AI-powered project suggestion engine (Claude API integration)
- Custom image upload — users supply their own pixel art source image
- Team / org graphs — contribution art for GitHub organization profiles
- Early adopter Lifetime Deal at $30-40 one-time payment

9. Monetization Strategy

9.1 Philosophy

PixelPush monetizes by enhancing the core experience — not gating it. The free tier is genuinely useful and shareable. Pro exists for developers who want a deeper, more personalized, and career-relevant experience. The goal is to make Pro feel like an obvious upgrade, not a paywall.

9.2 Free Tier

Feature	Free Tier
Image suggestions per theme	5 per theme
Active designs at once	1
Themes available	3 — Pets, Scenery, Abstract
Schedule recalculation	Once per active design
AI Project Suggestions	Not included
Portfolio Fit Score	Not included
Custom image upload	Not included
Org / team graphs	Not included
Social sharing	Enabled — with PixelPush watermark on share cards
Timelapse GIF	Not included
Support	Community / documentation only

9.3 PixelPush Pro — $6-8 / month

Pro is designed around two core upgrades: more creative options, and genuine career value. The AI Project Suggestion feature is the flagship Pro differentiator — it transforms PixelPush from a profile decoration tool into a portfolio-building companion.

Feature	Pro Tier
Image suggestions per theme	8 per theme (vs 5 on free)
Active designs at once	3 simultaneous designs
Themes available	All themes — current and future — plus early access
Schedule recalculation	Unlimited — full recovery from any missed days

AI Project Suggestions	3-5 tailored project ideas per design, matched to the user's detected tech stack and schedule duration
Portfolio Fit Score	Each project suggestion rated for interview and portfolio value
Custom image upload	Upload any image as the pixel art source
Org / team graphs	Contribution art for GitHub organization profiles
Social sharing	Watermark-free share cards and timelapse GIF export
Timelapse GIF	Auto-generated animated GIF of graph filling in over time
Support	Priority support with guaranteed response time

9.4 AI Project Suggestion Engine

The project suggestion feature is the most technically distinctive Pro capability. It uses the GitHub API data already collected during onboarding — specifically the user's public repositories — to infer their primary programming languages, frameworks, and project patterns. This data is combined with the selected design theme and schedule duration to generate highly personalized project recommendations via the Claude API.

How It Works

37. On design activation, PixelPush reads the user's public repos via GitHub API
38. Languages, frameworks, and project types are extracted and ranked by frequency
39. The design theme (e.g. Aviation) and schedule duration (e.g. 4 months) are noted
40. A structured prompt is sent to the Claude API with this context
41. Claude returns 3 to 5 project suggestions with titles, descriptions, estimated commit frequency, tech stack, and portfolio value rating
42. Suggestions are displayed on the Pro dashboard alongside the commit schedule

Example Output

User: Python developer · Theme: Aviation · Schedule: 4 months

Suggested Project: FlightPathCLI — A Python CLI tool that parses ADS-B flight data and visualises real-time aircraft trajectories using matplotlib. Estimated commits: 3-4 per week. Portfolio Value: High — demonstrates API integration, data parsing, and CLI design.

Why This Feature Justifies Pro

- Solves the ethical tension — users make real, meaningful commits instead of empty ones
- Adds career value independent of the graph art — users finish with a portfolio piece
- Highly personalized — no two users get the same suggestions

- Hard to replicate at free tier — AI inference cost is real, making it a natural Pro feature

9.5 Early Adopter Lifetime Deal

At launch, PixelPush will offer a one-time Lifetime Deal at $30-40, giving permanent Pro access for a single upfront payment. This strategy serves multiple purposes:

- Generates upfront revenue before subscription volume builds
- Creates a highly motivated cohort of early power users who advocate loudly
- Developers are conditioned to respond enthusiastically to lifetime deals — it is a proven launch tactic in the dev tools space
- Lifetime deal buyers become the primary source of early word-of-mouth and social sharing

Lifetime Deal Pricing Recommendation

$35 one-time — positioned as 'less than 5 months of Pro'. Cap at 200-500 redemptions to create urgency and prevent long-term revenue dilution.

9.6 Revenue Projections (Conservative)

Scenario	Revenue Estimate
500 users, 5% Pro conversion	~25 Pro users · $175-200/month MRR
2,000 users, 8% Pro conversion	~160 Pro users · $1,120-1,280/month MRR
10,000 users, 10% Pro conversion	~1,000 Pro users · $7,000-8,000/month MRR
Launch lifetime deal (200 sales)	$7,000 one-time · covers ~4 months of infrastructure

9.7 Deployment Cost vs. Revenue Crossover

Based on the Vercel + Neon + Resend infrastructure stack, PixelPush reaches profitability at approximately 9 to 10 Pro subscribers — an extremely low break-even threshold that is achievable within days of a successful launch post.

10. Security Considerations

10.1 OAuth Token Handling

- GitHub OAuth access tokens are encrypted at rest using AES-256
- Tokens are stored server-side only and never exposed to the client
- NextAuth.js handles session management with HTTP-only cookies

10.2 Rate Limiting

- GitHub GraphQL API allows 5,000 points per hour per authenticated user
- PixelPush batches API calls and caches results to stay well within limits
- Contribution data is synced once every 6 hours to minimize API usage

10.3 Data Privacy

- PixelPush only reads contribution metadata (dates, counts, colors) — never code or repository content
- Users may revoke GitHub access at any time from GitHub's application settings
- User data is permanently deleted within 30 days of account deletion

11. Glossary

Term	Definition
Contribution Graph	The 52-column x 7-row heatmap on a GitHub profile showing commit activity over the past year
Shade Level	One of 5 green intensity values GitHub uses to represent contribution density on any given day
Pixel Art Engine	PixelPush's internal system for converting theme images into 52x7 grid designs mapped to shade levels
Commit Schedule	The day-by-day plan generated by PixelPush specifying how many GitHub contributions to make on each date
Share Card	A dynamically generated PNG image (1200x630px) used for social media sharing of progress milestones
PWA	Progressive Web App — a web application installable on mobile and desktop with native-like capabilities
VAPID	Voluntary Application Server Identification — a protocol used to send Web Push notifications securely
Satori	A library developed by Vercel for converting JSX markup to SVG, used for server-side share card generation
AI Project Suggestion	A Pro feature that generates personalized GitHub project ideas matched to the user's tech stack and design theme using the Claude API
Portfolio Fit Score	A rating assigned to each AI-generated project suggestion indicating its value for software engineering interviews and portfolios
MRR	Monthly Recurring Revenue — total monthly income from active Pro subscriptions
Lifetime Deal	A one-time payment option offered at launch giving permanent Pro access, capped at a fixed number of redemptions
Serverless Function	A backend code execution model used by Vercel where functions run on-demand per request with no always-on server required

PixelPush — Product Documentation v1.1

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